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Microscopy-Based Modeling and Visualization of the Cell (Modeliranje in vizualizacija celice na osnovi mikroskopije)

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DOCTORAL DISSERTATION PROPOSAL

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Summary. Cell modeling is a critical field in biological research, providing insight into the complexity of cellular structures and functions. Within this doctoral thesis proposal, we propose the development of a statistical 3D model of eukaryotic cells and their internal organelles, integrating high-resolution data from Focused Ion Beam Scanning Electron Microscopy (FIB-SEM) and fluorescence microscopy. The methodology includes three stages: modeling organelle morphology using diffusion models with structural constraints, integrating these organelles into a unified whole-cell model, and developing a multi-scale visualization framework that transitions seamlessly between microscopy-like realism and illustrative abstraction. The resulting model will serve as a versatile tool for research, education, and science outreach, combining scientific accuracy with accessibility to engage diverse audiences.

1 Introduction

Cell modeling is becoming increasingly important in biological research, enabling scientists to understand the complexity of cellular structures and functions. The three-dimensional (3D) organization of cells and their internal organelles plays a fundamental role in cellular behavior [11]. Despite its importance, comprehensive models that accurately reflect this structural complexity remain limited. While individual cells exhibit considerable morphological diversity, statistical models, that capture shared structural features, can provide valuable insights into cellular organization and function.

Existing cell models often focus on isolated aspects of cellular architecture, such as nuclear morphology or overall cell shape [19, 37], without integrating detailed representations of multiple organelles and their spatial relationships. Many rely on conceptual rather than empirical data, limiting their effectiveness for quantitative research applications. Models based on lower-resolution microscopy techniques further restrict the level of detail available for analysis [10]. While conceptual models offer useful theoretical frameworks, they lack the empirical measurements necessary for quantitative studies.

Advances in microscopy techniques, including FIB-SEM and fluorescence microscopy, have generated detailed datasets of cellular ultrastructure. FIB-SEM provides nanometer-scale resolution for organelle morphology [49], while fluorescence microscopy captures overall cell shape and spatial distribution patterns [52]. Combining these complementary datasets enables the development of models that integrate fine structural details with broader spatial context.

The proposed thesis topic aims to develop a statistical 3D model of a eukaryotic cell and its internal organelles based on high-resolution microscopy data. The pipeline consists of three main stages: **(1)** modeling individual organelles to capture their morphology and variability; **(2)** integrating these organelle models into a unified whole-cell model representing both detailed organelle structures and their spatial organization; and **(3)** developing a visualization framework for multi-scale exploration of generated cells. The visualization framework will allow seamless transitions from whole-cell views (fluorescence microscopy resolution) to detailed views of

individual organelles (FIB-SEM resolution), aiding in understanding spatial relationships between organelles and their role in cellular processes.

The resulting model will serve as a valuable tool for research, education, and science communication. In research contexts, it can act as a reference framework for comparing cellular structures under different experimental conditions or identifying structural changes associated with diseases or treatments. Beyond research applications, the model supports educational efforts by representing cellular structures at different scales, making complex biological concepts accessible to students, educators, and the general public [12]. Unlike simplified textbook illustrations, these visualizations will be based on empirical data, combining scientific accuracy with accessibility to engage broader audiences.

2 Related work

Creating accurate cell models requires integrating multiple levels of structural detail. Existing frameworks such as CellOrganizer [29] typically focus on modeling individual cellular features independently, without explicitly representing spatial relationships among multiple organelles. Recent generative modeling approaches have demonstrated the potential of deep learning for capturing cellular organization in three dimensions. For instance, Sun et al. [43] introduced Generative Adversarial Network (GAN)-based methods trained on fluorescence microscopy data to generate realistic voxel-based predictions of subcellular organelles. However, their approach remains primarily voxel-based and does not explicitly represent organelles with continuous geometric representations such as implicit surfaces; additionally, it does not leverage higher-resolution imaging modalities like FIB-SEM.

Developing a comprehensive cell model involves several key steps: segmentation to delineate cellular structures (Section 2.1), modeling individual organelles to capture their morphology and variability (Section 2.2), and integrating these organelle models into a unified representation of cellular architecture (Section 2.3). Finally, visualization methods integrate all components into a seamless system (Section 2.4).

2.1 Segmentation

Accurate segmentation serves as the foundation for the model generation pipeline. While these segmentations can be produced manually or automatically, the large-scale data requirements of modern cell modeling make automated methods the preferred choice. Recent advancements in automated segmentation have significantly improved accuracy and efficiency in delineating cellular organelles. Specialized approaches tailored specifically to microscopy data and designed explicitly for whole-cell segmentation have been developed [5, 15, 25, 32, 33]. Several publicly available datasets support research in this area. The OpenOrganelle platform [49] provides pre-segmented organelle datasets primarily obtained from FIB-SEM imaging—the same modality used in the UroCell dataset [50, 51]. Annotated fluorescence microscopy datasets featuring cellular structures are also publicly accessible [3, 23, 28]. Raw microscopy data from repositories such as EMPIAR [18] can be segmented independently using publicly available models like Cellpose [42] or the frameworks mentioned above.

2.2 Single organelle modeling

The first step in the pipeline is modeling accurate organelle shapes to capture their morphological variability. FIB-SEM imaging, with isotropic voxel resolutions between 3 and 8 nm , provides an ideal solution for capturing the typical sizes of cellular organelles, which generally measure up to 2 μm .

Segmentations acquired from the micrographs often begin as voxel-based representations, but these can become memory-intensive at high resolutions and introduce aliasing artifacts [46]. To address these limitations, segmentations are converted into alternative representations such as implicit representations like Signed Distance Fields (SDFs) [35]. Implicit representations naturally handle complex topologies without requiring explicit connectivity information and effectively capture fine geometric details.

Deep learning techniques have significantly advanced shape modeling by directly learning continuous representations from data [44]. Neural implicit methods such as DeepSDF [36] and related approaches [6, 38] have demonstrated their effectiveness in capturing complex shape distributions. Recent advances in diffusion models have further improved the quality of 3D shape generation. Diffusion-SDF [39] applies this technique specifically to SDF representations by initially training on low-resolution grids and subsequently introducing finer details via patch-based super-resolution.

Structural constraints can be integrated into the diffusion pipeline to improve its applicability to organelle modeling. For example, regularization terms can enforce smooth curvature profiles observed in biological organelles [22], while loss functions incorporating bending energy minimization [43] ensure biologically plausible morphologies.

Recent approaches demonstrate effective methods for incorporating such constraints into diffusion processes. SCIGEN [34] applies constrained priors during denoising steps, ensuring adherence to predefined geometric patterns. Constrained Diffusion Models via Dual Training [20] optimize generative processes under statistical priors derived from empirical data, while Chung et al. [9] propose regularization techniques that improve geometric consistency during training.

By embedding biologically inspired constraints such as curvature regularization or bending energy minimization into diffusion model pipeline, organelle shape modeling can achieve higher fidelity while maintaining consistency with empirical observations of cellular architecture.

2.3 Cell organelle organization

Modeling the overall shape and spatial arrangement of organelles within a cell requires a different approach. Given an average cell size of approximately $30\ \mu m$, the high resolution provided by FIB-SEM imaging would result in excessively large datasets and is unnecessary for capturing general spatial arrangements. Instead, fluorescence microscopy, with resolutions typically around $200\text{--}250\ nm$, offers a more practical imaging modality.

Current approaches for modeling organelle distributions typically focus on individual organelle types independently. For instance, CellOrganizer employs Gaussian mixture models to represent vesicular organelle distributions [29]. While effective for single organelle classes, such methods fail to capture spatial interactions between different organelle types or enforce biologically relevant structural constraints (e.g., exclusivity among membrane-bound organelles).

Point process theory provides a theoretical foundation for spatial modeling. Simple Poisson processes represent random spatial distributions but cannot capture complex inter-organelle interactions. Multitype point processes extend this framework by incorporating interaction potentials that model attraction or repulsion patterns among multiple object classes [7]. Recent biological applications, such as CytoSpatio [4], have successfully employed these methods to learn spatial relationships across cellular environments.

However, statistical point processes have limitations in directly learning intricate spatial patterns from large datasets. Deep learning methods have significantly advanced spatial modeling capabilities by learning representations directly from data, and attention mechanisms have further improved spatial relationship modeling by selectively focusing on relevant features and capturing long-range dependencies [48].

Recently, Multi-Instance Diffusion (MIDI) [17] introduced a novel multi-instance attention mechanism specifically designed to capture inter-object interactions during the simultaneous generation of multiple 3D instances. Unlike previous methods that model objects independently or sequentially, MIDI explicitly captures inter-object interactions within the diffusion process, enabling accurate spatial relationships and efficient generation of multiple object classes. This makes MIDI particularly suitable for modeling organelle distributions within cells, where spatial coherence and inter-organelle interactions are critical.

Similarly to the single-organelle diffusion modeling step described earlier, MIDI’s applicability can be enhanced by integrating physically-inspired spatial constraints directly into its diffusion process. Attraction-repulsion dynamics provide a promising mechanism for enforcing structural constraints between generated instances.

Recent work by Christopher et al. [8] demonstrates how diffusion processes can be reformulated as constrained optimization problems, where projections at each step ensure adherence to specified constraints. Pair-wise potential functions between instances can be defined to enforce both attraction (ensuring proper clustering where appropriate) and repulsion (maintaining necessary separation between instances). These potentials can dynamically adjust during the sampling process, similar to methods in particle simulation systems [30], where distance-dependent forces regulate spatial arrangements.

By embedding physically-motivated constraints directly into MIDI’s multi-instance attention mechanism, generated distributions can satisfy both statistical realism and physical plausibility.

2.4 Visualization

Effective visualization techniques are essential for interpreting cellular models, whether for scientific analysis or educational purposes. Two main approaches are commonly used: microscopy-like visualizations, which replicate experimental imaging modalities, and illustrative visualizations, which abstract cellular models to emphasize specific features.

Microscopy-like visualizations replicate the appearance of experimental micrographs, allowing users to interpret geometric models in familiar contexts. Style transfer methods [14, 21] have been applied to transform geometric data into visually realistic microscopy-like images, while physical simulation tools [47] generate synthetic electron microscope images by modeling electron interactions with specimens. Advanced rendering techniques are then essential for efficiently visualizing the resulting high-resolution data. Adaptive multi-resolution encoding [45] and block-based volume rendering [40] optimize memory usage and ensure seamless

exploration by dynamically adjusting levels of detail. Herzberger et al. [16] introduced out-of-core rendering tailored for multi-channel microscopy data, while Lesar et al. [26, 27] evaluated depth-aware shading techniques to enhance spatial perception in crowded volumes. Nanotilus [2] addresses challenges in navigating crowded biological environments by employing sparsification techniques and guided camera path planning to reduce occlusion while preserving realism. Similarly, Nanomatrix [1] introduces scalable procedural scene construction strategies for visualizing large molecular datasets, enabling efficient transitions between atomistic detail and abstract representations.

Illustrative visualizations abstract cellular models to highlight specific features or relationships, making them effective for education and communication. Goodsell [12] pioneered molecular illustrations that simplify complex structures while maintaining scientific accuracy. Mindek et al. [31] combined realistic shading with illustrative highlights to communicate spatial relationships at different scales, while Sorger et al. [41] proposed abstraction transformations for transitioning between detailed molecular models and simplified representations.

Combining microscopy-like and illustrative visualizations has not been widely explored but is important for understanding cellular structures across scales. Microscopy Nodes in Blender [13] demonstrate how fluorescence microscopy data can be combined with segmented electron microscopy structures within the same view. Focus+context methods [24] overlay molecular trajectories onto mesoscopic cellular models, dynamically adjusting detail levels to guide attention in crowded environments. Integrating these styles into a unified framework requires adaptive rendering strategies that support seamless transitions between realism and abstraction while enabling multi-scale exploration.

3 Methodology

3.1 Cell organelle modeling

Segmented FIB-SEM microscopy data will be used as input to generate implicit shape representations of organelles. A deep diffusion model will be trained to generate statistically accurate shapes of organelles such as the nucleus, mitochondria, endolysosomes, fusiform vesicles, and Golgi apparatus. Structural constraints will be embedded in the training process to enforce biologically plausible properties such as smoothness, symmetry, and topology consistency. Quantitative evaluation of generated shapes will include metrics such as Chamfer distance, Intersection over Union (IoU), and surface smoothness.

3.2 Cell modeling

The overall cell shape and spatial arrangement of organelles will be modeled using a diffusion-based generative framework equipped with multi-instance attention mechanisms. Fluorescence microscopy data will provide spatial arrangement information, while the generated organelle shapes from the previous step will be positioned within the cell. Biologically inspired interaction potentials (e.g., attraction-repulsion forces) will be integrated into the attention mechanism to enforce realistic spatial constraints such as exclusivity among membrane-bound organelles. Spatial point process metrics (e.g., Ripley’s K-function) and expert biological assessments will be used for validation.

3.3 Multi-scale interactive visualization

A unified visualization framework will be developed to enable seamless exploration across scales (whole-cell views to individual organelle details) and styles (microscopy-like realism to illustrative abstraction). The framework will:

- Implement interactive multi-scale navigation techniques for intuitive exploration of cellular structures.
- Develop style-transfer methods for generating microscopy-like visualizations from statistical models and transitions to abstract styles.
- Provide interactive controls for dynamically switching between visualization styles.

Evaluation will involve expert feedback sessions and structured user studies assessing clarity in conveying complex structural relationships using synthetic and real microscopy datasets.

4 Expected contributions to science

1. **A deep diffusion-based approach for generating morphologically accurate models of key organelles** such as mitochondria, endolysosomes, fusiform vesicles, and Golgi apparatus. Unlike existing methods that focus on individual organelle types or lack statistical modeling, this approach integrates high-resolution microscopy data with structural constraints to ensure biological plausibility.

2. **A generative model that captures overall cell shape and spatial organization of organelles** using fluorescence microscopy data. This method explicitly models inter-organelle interactions through biologically inspired potentials, addressing limitations in current approaches that neglect spatial relationships or oversimplify cellular architecture.
3. **A novel visualization technique enabling seamless transitions across scales** (whole-cell views to individual organelle details) **and styles** (microscopy-like realism to illustrative abstraction). By integrating physically-based rendering with style-transfer methods, the proposed approach improves usability for scientific analysis and educational communication while addressing gaps in existing biological visualization tools.

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